# 2021 HW1

MOS Theory, Multiplexers, and Static CMOS

EE3032 / Introduction to VLSI Design / Jin-Fu Li

**Provenance.** The original assignment and the supplied answers by Lin Yan-Yu were read first. The results below were independently derived, with the useful reference choices retained and improvements identified. Source details and hashes are in the year-level `SOURCES.md`.

**Notation.** + is OR; juxtaposition is AND. Every pMOS body is tied to $V_{\mathrm{DD}}$ and every nMOS body to ground. Body terminals are omitted in the diagrams. Only true input literals are used in the CMOS gates below.

## 1. Two identical series devices per inverter branch

**(a) The rail-side devices $M_1$ and $M_4$ cannot carry nonzero DC current in saturation.** For the nMOS stack let the internal node be $u$ and $U = V_{\mathrm{in}} - V_{tn} > 0$. Saturation of the lower $M_1$ would require $u \geq U$. But then the upper device has $V_{GS,2} = V_{\mathrm{in}} - u \leq V_{tn}$ and is cut off. Thus the assumed positive stack current is impossible. The pMOS argument is identical with voltages measured downward from $V_{\mathrm{DD}}$: saturation of rail-side $M_4$ would cut off $M_3$. At cutoff or the zero-current boundary, the usual region inequalities may coincide; the statement concerns conducting operation. **(b) Both equivalent aspect ratios are one half of the original:**

$$\boxed{(W/L)_{N_1} = (W/L)_{N_2} = \tfrac{1}{2}(W/L).}$$

Here $N_1$ is nMOS and $N_2$ is pMOS. This equality uses the ideal long-channel square-law model, no body effect, and no channel-length modulation; it does not assert exact BSIM short-channel equivalence. Let $k_n = \mu_n C_{\mathrm{ox}}(W/L)$. If the upper nMOS is saturated,

$$k_n(Uu - u^2/2) = \frac{k_n}{2}(U-u)^2, \quad u = U\left(1 - \frac{1}{\sqrt{2}}\right), \quad I = \frac{k_n U^2}{4}.$$

A single saturated nMOS with $k_{\mathrm{eq}} = k_n/2$ has the same current. If both devices are linear and $v = V_{\mathrm{out}} < U$, adding their equal-current equations gives

$$\frac{2I}{k_n} = Uv - \frac{v^2}{2}, \qquad I = \frac{k_n}{2}\left(Uv - \frac{v^2}{2}\right).$$

Thus the half-aspect-ratio device also matches the linear part of the branch characteristic; the pMOS derivation is symmetric. Halve $W$ at fixed $L$, or double $L$ in this ideal model. **Reference comparison.** The supplied answer identifies $M_1, M_4$ and the factor $1/2$ correctly. The present proof also covers the linear region. `verify_logic.py` checks 5,050 normalized operating points; the maximum current-equation error is below $10^{-12}$.

## 2. Exactly two of three inputs are one

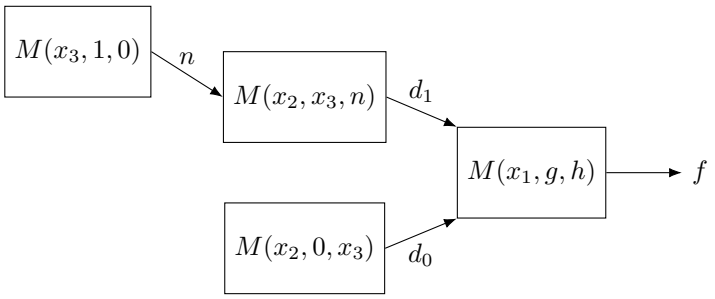
The requested function is

$$f = \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 = \bar{x}_1 (x_2 x_3) + x_1 (x_2 \oplus x_3).$$

Write $M(s, d_0, d_1) = \bar{s} d_0 + s d_1$. An implementation using four 2:1 multiplexers and constants only is

$$n = M(x_3, 1, 0), \quad g = M(x_2, 0, x_3), \quad h = M(x_2, x_3, n), \quad \boxed{f = M(x_1, g, h).}$$

The first mux makes $\bar{x}_3$ without an inverter; $g = x_2 x_3$ and $h = x_2 \oplus x_3$.




| $x_1$ | $x_2$ | $x_3$ | $f$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

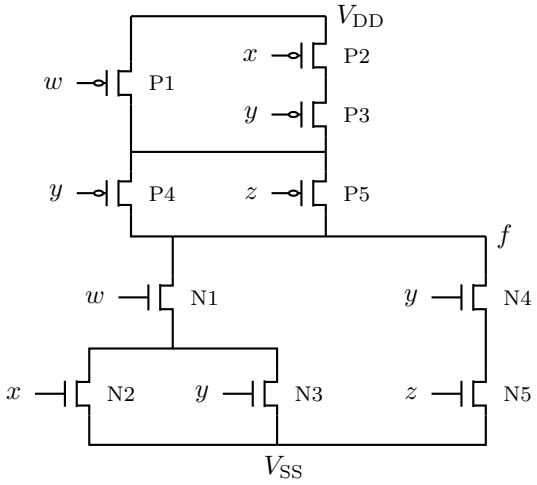
**Reference comparison.** The supplied full Shannon tree has seven muxes, including constant-only leaves. Its truth table is correct. The factored four-mux circuit saves three muxes while still using muxes exclusively. Exhaustive evaluation of all eight inputs confirms equality.

# 3(a). Complemented AOI function

**Read the overbar in the prompt carefully: it spans the entire sum.**

$$\boxed{f(w,x,y,z) = \overline{w(x+y)+yz}.}$$

The pull-down network (PDN) implements $G = w(x+y)+yz$: $n_w$ in series with $(n_x \parallel n_y)$, in parallel with ($n_y$ series $n_z$). The pull-up network (PUN) is the dual: $(p_w \parallel (p_x \text{ series } p_y))$ in series with $(p_y \parallel p_z)$.




There are five nMOS and five pMOS devices: **10 transistors total**. Two devices are separately controlled by $y$ in each network; they are distinct transistors.

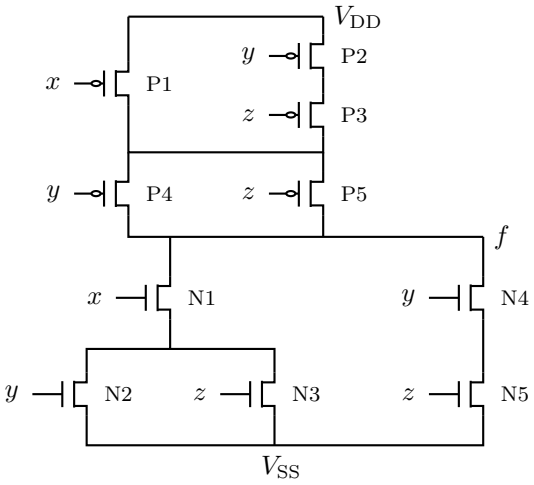
$$PUN = [\bar{w} + \bar{x}\bar{y}](\bar{y} + \bar{z}) = \overline{w(x+y)+yz} = f.$$

Every input has exactly one conducting rail network in ideal switch logic, so there is no static VDD-to-ground path. No complemented external input is needed. **Reference comparison.** The supplied diagram uses the same correct 10-transistor topology. This solution retains it, clarifies the overbar, and checks all 16 input combinations.

## 3(b). Complement of a three-input majority

$$f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z} = \overline{xy + xz + yz} = \boxed{\overline{x(y+z) + yz}}.$$

Use the same network structure as 3(a), now with the first branch $n_x$ in series with $(n_y \parallel n_z)$ and the second branch $n_y$ in series with $n_z$.




**Gate labels in this diagram.** P4/N4 are controlled by $y$ and P5/N5 by $z$; repeated labels denote independent devices. The dual PUN is $(p_x \parallel (p_y \text{ series } p_z))$ in series with $(p_y \parallel p_z)$. It uses true literals only and needs **10 transistors**.

| $x$ | $y$ | $z$ | $f$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

**Reference comparison.** The supplied answer implements the equivalent PDN $(x+y)(x+z)(y+z) = xy + xz + yz$, using six nMOS and six pMOS. Factoring the majority expression saves two transistors. All eight combinations pass the independent PDN/PUN complement check.